

Álvaro Arturo Sanjuán Cuéllar

Curriculum Vitae

Personal Info

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Research Interests

Nonlinear analysis applied to partial differential equations: the wave equation, elliptic boundary value problems, singular fourth-order ordinary differential equations (nanobeam and MEMS models), and delay functional-differential equations, using topological methods (Leray-Schauder degree) and numerical methods (shooting methods, finite elements, spectral/FFT techniques).

Education

1996–2002 **(BS) Mathematician**, *Universidad Nacional de Colombia*, Bogotá
(Matemático)

- 2003–2005 **(MS) Master in Mathematical Sciences**, *Universidad Nacional de Colombia*, Bogotá
(Magister en Ciencias-Matemáticas)
- 2009–2015 **(PhD) Mathematical Sciences**, *Universidad Nacional de Colombia*, Bogotá
(PhD Ciencias-Matemáticas)

Research Stays

feb-may 2018, Buenos Aires Argentina

- subject *Functional differential equations applied to biology*
- advisor Pablo Amster (PhD)
- institutions Universidad de Buenos Aires (UBA), Centro Latinoamericano de Formación Interdisciplinaria (CELF)
- may-aug 2018, Madrid Spain
- subject *Solutions to superlinear elliptic problems*
- advisor Rosa Pardo (PhD)
- institution Universidad Complutense de Madrid (UCM)

PhD Dissertation

- title *Vibrating Membranes*
- advisor José Francisco Caicedo, Alfonso Castro
- description Bifurcation at infinity for the semilinear wave equation $\square u + \lambda u + h(u) = 0$ was proven.

Academic Distinctions

- 2016 Honouring mention – Yu Takeuchi medal for best works in mathematical sciences. *Colombian Academy of Exact, Physical and Natural Sciences (Academia Colombiana de Ciencias Exactas, Físicas y Naturales)*.

Experience

Teaching

- 2003–2006 **Professor**, *Universidad de los Andes*, Bogotá
Professor of Mathematics.
- 2006–current **Professor (Profesor Titular)**, *Universidad Distrital Francisco José de Caldas*, Bogotá
Professor of Mathematics.
- Courses Mathematical Analysis, Differential Equations, Numerical Analysis, Surface Geometry, Calculus of Variations, Topology.

Research

- 2016–2017 **Principal researcher "Weak solutions to the semilinear wave equation"**, *Universidad Distrital Francisco José de Caldas*, Bogotá

2025–2026 **Principal researcher "Euler-Bernoulli Equations in Nanobeam Models" (project PIS-2025-2063), Universidad Distrital Francisco José de Caldas, Bogotá**

Academic Service

- Reviewer American Mathematical Society (AMS).
Secretary Sociedad Colombiana de Matemática (SCM).
Director Semillero en Análisis Matemático (SAMAT), Universidad Distrital Francisco José de Caldas. Research seminar group for undergraduate student development; organized the I Encuentro Juvenil en Ecuaciones Diferenciales (April 2025).

Speaker at Professional Meetings

- oct 2010 Mathematical Physics Seminar, Bologna University - Italy. *Periodic Solutions of a Semilinear Wave Equation in Higher Dimensions.*
feb 2017 ICMC Summer Meeting on Differential Equations 2017 Chapter – San Carlos, San Pablo Brazil. *Bifurcations from infinity for a semilinear wave equation.*
may 2021 UNC Greensboro Partial Differential Equations Conference – UNC Greensboro A *priori* bounds for radial solutions to elliptic equations approaching critical growth.
sep 2021 VI Latinamerican Congress of Mathematics – Montevideo, Uruguay. *A Lazer-Leach type result for functional-differential equations at resonance.*
jun 2026 IV Colombian Conference on Industrial and Applied Mathematics (MAPI 4), Universidad Icesi, Cali, Colombia. *Existence and Computation of Periodic Solutions for the Semilinear Gompertz Equation with Discrete Delay.*
oct 2025 3rd UNCG Virtual PDE Conference. *The Work of Alfonso Castro on the Semilinear Wave Equation: A Survey.*

Selected Publications

- 2014 with: Caicedo, J. F., Castro, A. and Duque, R. *Existence of solutions for a semilinear wave equation with non-monotone nonlinearity* In: Discrete and Continuous Dynamical Systems – S 7(69) pp. 1193-1202.
2017 with: Caicedo, J. F. and Castro, A. *Bifurcation at Infinity for a Semilinear Wave Equation* In: Discrete and Continuous Dynamical Systems – A 37 (4) pp. 1857 - 1865
2017 with Caicedo, J. F., Castro, A. and Duque, R. *The Semilinear Wave Equation with Non-monotone Nonlinearity: A Survey* In: Rend. Istit. Mat. Univ. Trieste (49) pp. 207-214.
2020 with Pardo, R. *Asymptotic behavior of positive radial solutions to elliptic equations approaching critical growth.* In: Electronic Journal of Differential Equations fasc. 114 pp. 1-17
2021 with Amster, P. and Epstein, J. *Periodic solutions for systems of functional-differential semilinear equations at resonance.* In: Topological Methods in Nonlinear Analysis. Juliusz Schauder Center (58) fasc. 2 pp. 591–607

- 2026 *Existence and Computation of Periodic Solutions for the Semilinear Gompertz Equation with Discrete Delay*. In: Proceedings of the Fourth Colombian Conference on Applied and Industrial Mathematics (MAPI 4). Springer. (Accepted)
- 2026 with Rodriguez, F. *A System of Semilinear Wave Equations with Non-monotone Nonlinearity*. Dedicated to Professor Alfonso Castro. Submitted to the Electronic Journal of Differential Equations (EJDE), Special Issue in Honor of Alfonso Castro (UNC Greensboro PDE Conference, 2025).

Graduate Student Supervision

- 2019 Master's Thesis: Fernando Rodriguez, *A System of Semilinear Wave Equations with Non-monotone Nonlinearity*. Universidad Nacional de Colombia – Sede Bogotá.

Languages

Spanish	Native speaker	
English	TOEFL iBT 2017	B2, 88
	IELTS ACADEMIC 2023	B2, 6.5

Programming languages

C++, Python, R, Sage, Mathematica, Julia, Maxima